Rainbow perfect matchings on even cycles and two disjoint squares

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Abstract. We determine the maximum number of colors in an edge coloring of an even cycle that avoids a rainbow perfect matching and count all extremal partitions of its fixed edge set. For two disjoint squares, we prove that the maximum is six and that exactly two extremal partitions occur. The arguments are elementary and self-contained; no novelty is claimed.

1. Definitions and the even-cycle formula

All graphs are finite and simple. Edge colorings need not be proper. A perfect matching is *rainbow* when its edges have pairwise distinct colors. Let $A(G)$ be the largest number of colors in a coloring avoiding every rainbow perfect matching. Let $N_{\mathrm{ext}}(G)$ count the extremal partitions of the fixed edge set into color classes. Color names are ignored; vertex symmetries are not identified.

Theorem 1. For every integer $n \geq 2$,

$$A(C_{2n}) = 2n - 2, \quad N_{\mathrm{ext}}(C_{2n}) = [n(n-1)/2]^2.$$

Proof. The cycle has exactly two perfect matchings, M_1 and M_2: one choice at a vertex forces the alternating edges around the cycle. These matchings partition its $2n$ edges. Let S_i be the palette on M_i. Each palette has at most $n - 1$ colors, so

$$|S_1 \cup S_2| \leq |S_1| + |S_2| \leq 2n - 2.$$

For equality, merge one pair of edges in each matching into a color class and give all remaining edges distinct colors, with disjoint palettes between the matchings. This uses $2n - 2$ colors and avoids a rainbow matching.

Conversely, equality forces both palettes to have size $n - 1$ and to be disjoint. A partition of n edges into $n - 1$ classes has exactly one doubleton and otherwise singletons. Choosing that pair independently in each matching gives $[n(n-1)/2]^2$ distinct partitions, and every extremizer arises uniquely this way. ■

2. The maximum for two disjoint squares

Theorem 2. If G is the disjoint union of two copies H_1, H_2 of C_4, then $A(G) = 6$.

Proof. Color the edges of H_1 in cyclic order A, B, A, B. Give the four edges of H_2 four distinct new colors. Every global perfect matching contains one of the two monochromatic local matchings in H_1. Thus six colors are attainable without a rainbow matching.



Figure 1. An avoiding six-coloring; the first square forces a repeated color.

For the upper bound, consider a coloring using at least seven colors on the eight edges. With eight colors, every perfect matching is rainbow. With seven colors, precisely one pair e, f shares a color and all other classes are singletons. In the component containing e, choose the alternating matching omitting e. Combine it with either matching of the other component. The resulting perfect matching cannot contain the only repeated pair, so it is rainbow. This proves the upper bound. ■

3. Characterizing all extremal partitions

Theorem 3. With edge positions fixed and color names ignored, $N_{\mathrm{ext}}(G) = 2$ for the graph of Theorem 2.

Proof. Six nonempty color classes on eight edges have total excess size two over singletons. Their sizes are therefore either $(3, 1, 1, 1, 1, 1)$ or $(2, 2, 1, 1, 1, 1)$.

Excluding the triple. There are four global perfect matchings, formed by independently choosing one of two local matchings in each component. Every edge belongs to exactly two of them. Three equal-colored edges therefore have six total appearances across the four matchings. If every matching were non-rainbow, each would contain at least two of these edges, requiring at least eight appearances. This contradiction rules out a triple.

3. Characterizing all extremal partitions (continued)

Two repeated pairs. Let P and Q be the two doubleton color classes. A matching is non-rainbow exactly when it contains P or Q in full. A pair is contained in the following number of global perfect matchings:

Pair position	Number of matchings
Adjacent edges in one component	0
Opposite edges in one component	2
One edge in each component	1

Adjacent edges cannot occur together. Opposite edges specify one local matching, leaving two choices in the other component. One edge in each component specifies both local choices and hence exactly one global matching.

Together, the two repeated pairs must obstruct all four matchings. Each obstructs at most two, so both must obstruct exactly two, and the obstructed sets must be disjoint. Thus each pair consists of opposite edges within one component. If the pairs belonged to different components, choosing the other local matching in each component would give a rainbow global matching.

Both pairs must therefore lie in the same component. Because the color classes are disjoint, they are precisely its two alternating perfect matchings. There are exactly two choices for that component; the other component has four singleton classes. Each choice avoids a rainbow matching, and the choices define different fixed-edge partitions. Renaming or interchanging colors adds no further partitions. Thus the count is two. ■

4. Small cases and computational checks

Host	Maximum colors	Extremal partitions
C_4	2	1
C_6	4	9
C_8	6	36
C_{10}	8	100
Two disjoint C_4s	6	2

All entries follow from the proofs. The repository's documented sparse search also checks the three cycles through eight edges and the disjoint squares, agreeing with these formulas. The C_{10} row is a proof consequence; no additional computation is claimed here. Computation is not a premise of any theorem in this note.

For the edge order, raw partitions, reproduction commands, and execution limits, see Discrete Math Lab, PR #1 and the files *results/data/sparse_rainbow.json* and *results/sparse_verification.md*. The counts concern color partitions on fixed edges, not graph isomorphism classes.